



DAY 2 • AYE TECH HUB

How AI Works

Day 2 — Under the Hood: Neural Networks, Training, and Why GPUs Matter

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Table of Contents

About This Lesson	3
1. The Big Idea — AI = Patterns from Data	3
2. The Three Stages of Every AI System	4
3. What Is a Neural Network?	5
4. How Training Actually Works	6
5. Parameters, Weights, and Why Size Matters	7
6. Why GPUs Power Modern AI	8
7. A Worked Example — Recognising a Cat	9
8. Practical Exercises	10
9. Self-Check Quiz	11
Summary & Tomorrow's Lesson	12

About This Lesson

Yesterday on **Day 1** you learned *what* Artificial Intelligence is. Today on **Day 2** we open the hood and look at *how* it actually works. By the end of this lesson, terms like **neural network**, **training**, **weights**, and **GPU** will stop sounding mysterious — they'll feel like ordinary engineering vocabulary.

We will keep this visual and practical. No advanced maths is required. If you can multiply two numbers, you already have all the maths you need to follow today's lesson.

■ Learning Outcomes

By the end of today you will be able to: **(a)** describe the three universal stages of any AI system (data → training → inference); **(b)** draw a simple neural network in your notebook; **(c)** explain in plain words what "training a model" actually means; **(d)** say what weights and parameters are; **(e)** explain why GPUs — not CPUs — are the engine of modern AI.

1. The Big Idea — AI = Patterns from Data

Here is the single most important sentence in this whole course:

■ The One-Sentence Definition

Modern AI is software that finds patterns in huge amounts of data and uses those patterns to make predictions on new data it has never seen before.

Everything else — neural networks, GPUs, transformers, LLMs — is engineering detail in service of that one idea.

Think about how a child learns to recognise a cat. Nobody hands the child a rulebook saying "a cat has pointed ears, whiskers, and four legs." Instead, the child sees hundreds of cats — different colours, different angles, different lighting — and slowly builds an internal sense of "cat-ness." Modern AI works the same way: **show it enough examples and it figures out the pattern on its own.**

2. The Three Stages of Every AI System

Every AI system you'll ever build, buy, or use goes through the same three stages. Memorising these gives you a mental map for the entire field:

1

Stage 1 — Data Collection & Preparation

Gather thousands or millions of examples. Clean them, label them, and split them into training and testing sets. Garbage in = garbage out, so 60-80% of real-world AI work happens here.

2

Stage 2 — Training the Model

Feed the data into an algorithm (usually a neural network). The algorithm adjusts its internal numbers thousands of times until it gets good at predicting the right answer. This is where the GPU sweat happens.

3

Stage 3 — Inference (Using the Model)

The trained model is now deployed. It receives new, unseen input and produces a prediction or output. Every time you type into ChatGPT or your phone unlocks with your face, you're running inference.

■ Mental Shortcut

Training = teaching the model (slow, expensive, done once). **Inference** = using the model (fast, cheap, done billions of times). The same trained model can serve millions of users.

3. What Is a Neural Network?

A **neural network** is the algorithm at the heart of modern AI. The name is biological but the maths is simple: it is a stack of layers of tiny mathematical units called **neurons** (or **nodes**). Each neuron does a small calculation and passes its result to the neurons in the next layer.

Think of a neural network as a series of filters:

- **Input Layer** — receives the raw data (e.g. pixels of an image, words of a sentence).
- **Hidden Layers** — do the actual pattern-finding. Modern networks can have anywhere from 10 to over 100 hidden layers — this is where the word "**deep** learning" comes from.
- **Output Layer** — produces the final prediction (e.g. "this is a cat" or the next word in a sentence).

Each connection between neurons has a **weight** — a number that says how much that connection matters. A simple network for recognising handwritten digits might have **about 100,000 weights**. A modern Large Language Model like the one behind ChatGPT has **hundreds of billions**.

■ Industrial Analogy

If you are familiar with industrial control, a neural network is like a giant PLC ladder logic — except instead of an engineer writing each rung by hand, the rungs are learned automatically from data, and there can be billions of them.

4. How Training Actually Works

Training is where the magic happens — but it's not magic, it's a loop of guess-and-correct repeated millions of times. Here is the full cycle, in plain English:

1

Forward Pass

Show the network one training example (e.g. an image of a cat) and let it produce a guess at the answer.

2

Compare to the Correct Answer

Measure how wrong the guess was. This number is called the **loss** or **error**. A perfect guess has loss = 0.

- 3 Backward Pass (Backpropagation)**
Trace the error backwards through the network and figure out which weights contributed most to the mistake.
- 4 Update the Weights**
Nudge each weight slightly in the direction that would have reduced the error. The nudge size is called the **learning rate**.
- 5 Repeat — Millions of Times**
Cycle through the whole training set many times (each full pass is called an **epoch**). After enough epochs, the network's guesses become highly accurate.

■ Training Is Expensive

Training a state-of-the-art LLM in 2026 can cost **tens of millions of dollars** in electricity and GPU rental — and consume as much power as a small town for weeks. This is why very few companies train frontier models from scratch.

5. Parameters, Weights, and Why Size Matters

When you read about "a 70-billion parameter model," what does that number actually mean? It is simply the count of **tunable numbers** (weights and biases) inside the neural network. The more parameters, the more patterns the model can potentially capture — but the more data and computation it needs to train.

Era	Typical Parameter Count	Example
Early ML (2000s)	~10,000	Spam filters
Image AI (2012)	~60 million	AlexNet
Early LLMs (2018)	~110 million	BERT-base
Mid-Era LLMs (2020)	~175 billion	GPT-3
Modern Open LLMs (2024)	~70 billion	Llama 3 70B
Frontier LLMs (2025-26)	~1+ trillion (est.)	GPT-5, Claude Opus 4

■ Bigger Isn't Always Better

A small, well-trained model on the right data often beats a huge model trained on junk. For your own AYE Tech Hub projects, a **7B-parameter open model** running on a single GPU is often the sweet spot — affordable, fast, and good enough.

6. Why GPUs Power Modern AI

A normal computer processor (CPU) is like a brilliant scholar who can solve any problem — one at a time, very quickly. A graphics processor (GPU) is like a stadium full of school

children, each doing simple multiplications in parallel. Neural networks happen to be billions of simple multiplications, so the stadium wins.

	CPU	GPU
Cores	4 – 64	5,000 – 20,000+
Each core	Very fast, very smart	Slower, simpler
Best at	Sequential tasks	Massive parallel maths
AI training	Too slow	10× to 1000× faster
Famous brand	Intel, AMD	NVIDIA (H100, B200)

This is why NVIDIA became one of the most valuable companies on Earth: their GPUs are the picks-and-shovels of the AI gold rush. Beyond GPUs, specialised AI chips — Google's **TPUs**, AWS's **Trainium**, Apple's **Neural Engine** — push the same idea even further.

7. A Worked Example — Recognising a Cat

Let's walk through a complete (simplified) AI system end-to-end. Goal: a system that decides whether a photo contains a cat.

- 1 Collect Data**
Gather 10,000 photos: 5,000 labelled "cat" and 5,000 labelled "not a cat." Split into 8,000 for training and 2,000 for testing.
- 2 Choose an Architecture**
Pick a Convolutional Neural Network (CNN) — a neural network specialised for images. Say it has 5 million parameters.
- 3 Train on the GPU**
Run the training loop from Section 4 for ~50 epochs. On a single modern GPU this takes 30 minutes to a few hours.
- 4 Evaluate on Unseen Photos**
Test on the 2,000 photos the model has never seen. If accuracy is, say, 96%, you have a usable model.
- 5 Deploy for Inference**
Save the trained weights to a 20 MB file. Now any phone, server, or website can load it and classify new photos in milliseconds — no GPU needed for inference.

■ Same Recipe, Different Problems

Swap "photos of cats" for any other data and the recipe stays the same: **chest X-rays** → detect pneumonia; **vibration sensor data** → predict bearing failure in a motor; **customer reviews** → flag negative feedback automatically. The pattern is universal.

8. Practical Exercises

Spend 20–40 minutes on these today. Hands-on practice is what moves AI from "interesting" to "useful" in your work.

Exercise 1 — Sketch a Neural Network (10 min)

Get a sheet of paper. Draw 3 circles labelled "Input" on the left, 4 circles labelled "Hidden" in the middle, and 2 circles labelled "Output" on the right. Connect every input circle to every hidden circle with a line, and every hidden circle to every output circle. **Count the lines.** Each line is a weight. You should get 20. Congratulations — you just designed a 20-parameter neural network.

Exercise 2 — Try a Real Trained Model (15 min)

Open **Teachable Machine** in your browser (teachablemachine.withgoogle.com). It is free, no installation required. Train a simple image classifier in 3 minutes:

1. Click "Get Started" → "Image Project" → "Standard".
2. Use your webcam to record ~30 images of yourself holding nothing.
3. Record ~30 images of yourself holding any object (a pen, a phone).
4. Hit **Train Model** and watch the loss decrease in real time.
5. Test it live — wave the object in and out of view and see the predictions flip. You just did the full "data → training → inference" cycle.

Exercise 3 — Ask an LLM About Itself (10 min)

Open ChatGPT, Claude, or Gemini and ask these three questions:

1. "In one paragraph, explain how you were trained."
2. "Roughly how many parameters do you have? What does that mean?"
3. "When you answer me, are you being trained, or doing inference? Explain the difference."

Exercise 4 — Apply It to Your Field (5 min)

Write 2–3 sentences in your notebook describing one specific task in your work or studies that could be solved with the "data → training → inference" pattern. Example: *"Predict when a specific robot needs maintenance, by training on its vibration and current-draw history."*

9. Self-Check Quiz

Try to answer without scrolling back. Answer key follows.

1. Name the three universal stages of an AI system.
2. Inside a neural network, what is a **weight**?
3. What is the difference between **training** and **inference**?
4. What is one full pass through the training data called?
5. Why are GPUs much better than CPUs for training neural networks?

6. True or False: A model with more parameters is always smarter.
7. A model is trained, saved as a file, then loaded on a phone to classify photos. Is the phone doing training or inference?

■ Answer Key

1. Data preparation, training, inference. 2. A tunable number on a connection between two neurons; the trained model is essentially a giant collection of weights. 3. Training teaches the model on data (slow, expensive); inference uses the trained model on new input (fast, cheap). 4. An epoch. 5. GPUs have thousands of simple parallel cores, perfect for the millions of simultaneous multiplications a neural network performs. 6. False — quality of data and architecture matter more than raw size for most real-world problems. 7. Inference.

Summary & Tomorrow's Lesson

Concept	Essential Point
AI in one line	Software that learns patterns from data and predicts on new data
Three stages	Data preparation → Training → Inference
Neural network	Layers of neurons connected by weights; "deep" means many layers
Training loop	Guess → measure error → backpropagate → update weights → repeat
Parameters	Count of tunable numbers in the model; bigger = more capacity but more cost
Why GPUs	Massive parallel maths; thousands of cores vs CPU's dozens
Inference	Using the trained model — fast, cheap, runs everywhere

■ Coming Up — Day 3: AI in Daily Life

Tomorrow we zoom back out and look at where AI is already living around you — in your phone, your car, your bank, your hospital, your factory. You'll start seeing AI everywhere once you know the patterns to spot.

■ Continue Your AI Journey with AYE Tech Hub

This guide is Day 2 of our **30-Day AI Course** under **AYE Tech Hub**. Continue with the full series for PLC, automation, CAD/Revit, electrical, and mechanical engineering content. Also explore PDFs, AI tools and ebooks on **AYE TechPro**.

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Day 2 of the AYE Tech Hub 30-Day AI Course.

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